

PAPER

Machine learning-enabled prediction of optical emission spectra in scalable non-thermal Ar atmospheric pressure plasma multi jet

To cite this article: P S N S R Srikar *et al* 2025 *J. Phys. D: Appl. Phys.* **58** 415204

View the [article online](#) for updates and enhancements.

You may also like

- [Autler–Townes splitting in Rydberg atoms: transition dipole matrix element extraction and field efficiency analysis](#)
Brian C Holloway, Gavin M Chase, Lee E Harrell et al.
- [ICRH modelling of DTT in full power and reduced-field plasma scenarios using full wave codes](#)
A Cardinali, C Castaldo, F Napoli et al.
- [Global evidence that cold rocky landforms support icy springs in warming mountains](#)
Stefano Brighenti, Constance I Millar, Scott Hotaling et al.

Machine learning-enabled prediction of optical emission spectra in scalable non-thermal Ar atmospheric pressure plasma multi jet

P S N S R Srikar , Indhu Suresh  and R K Gangwar* 

Department of Physics & CAMOST, Indian Institute of Technology Tirupati, Yerpedu 517619, India

E-mail: reetesh@iittp.ac.in

Received 23 June 2025, revised 1 September 2025

Accepted for publication 25 September 2025

Published 8 October 2025



Abstract

In this study, an Ar-multi-non-thermal atmospheric pressure plasma jet system was designed and fabricated to explore its scalability for large-volume processing. Optical emission spectroscopy measurements were conducted as a function of various operating parameters, including gas flow rate, voltage, duty cycle, and % air. The measurements revealed strong emissions from Ar I, originating from fine-structure transitions among the $3p^5 4p \rightarrow 3p^5 4s$ levels within the 690–810 nm wavelength range. Additionally, significant emissions were observed from the OH rotational band ($A^2\Sigma^+ \rightarrow X^2\Pi$) at 309 nm, the N_2 second positive system ($C^3\Pi_u \rightarrow B^3\Pi_g$) between 330 nm and 380 nm, and atomic oxygen ($2s^2 2p^3 ({}^4S^0) 3p^5 P \rightarrow 2s^2 2p^3 ({}^4S^0) 3s^5 S^0$) at 777.5 nm, providing valuable insights into reactive chemical species formation. To analyze the influence of operating conditions on these emissions and their impact on reactive species production, a data-driven machine learning (ML) approach was employed, leveraging a combination of dimensionality reduction and supervised learning. Principal component analysis was first applied to reduce data dimensionality, enhancing computational efficiency while preserving essential spectral features. Subsequently, two supervised ML models, Random Forest and deep neural networks, were trained to map input conditions to predicted emission intensities. To further exploit the predictive capability of these models, a differential evolution optimization strategy was implemented to estimate the optimal combinations of input parameters that maximize specific spectral features, particularly OH(A–X) emission. This study contributes to the development of more efficient and accelerated plasma diagnostic approaches, with the potential for real-time predictions.

Keywords: machine learning, plasma diagnostics, optical emission spectroscopy, nonthermal plasma

1. Introduction

Atmospheric pressure low-temperature plasmas are increasingly recognized for their vast potential in large-volume processing applications, owing to their ability to generate reactive

and energetic species under ambient conditions. Among these, argon-based non-thermal atmospheric pressure plasma jets (Ar-NTAPPJs) have emerged as promising, with extensive use in fields ranging from material processing and medicine to energy and environmental applications [1]. Scaling these plasma systems promises to amplify the production of reactive species, enabling continuous and large-volume processing. However, this scaling introduces complex technical

* Author to whom any correspondence should be addressed.

and operational challenges that must be addressed to harness their capabilities thoroughly.

One of the primary hurdles in scaling NTAPPJs is generating an adequate concentration of reactive species over the scaled plasma system in the ambient environment. While single-jet systems allow for relatively controlled spatial distributions of radicals, ions, and electrons, increasing the number of jets leads to intricate interactions between individual plasma streams. These interactions can cause inconsistencies and higher gradients of reactive species across plasma interaction surfaces, potentially resulting in adverse effects in applications where uniform exposure is critical, such as surface modification [2]. Furthermore, larger systems necessitate higher power inputs, complicating power management and cooling strategies [3]. Operating multiple plasma jets simultaneously can introduce interference between jets, disrupting the overall stability of the system [4]. The complexity is further compounded by an increased number of operational variables—such as gas flow rates, voltage, and duty cycle—that make real-time control and optimization more challenging [2]. In this regard, non-invasive optical emission spectroscopy (OES)-based diagnostic approaches are widely used. However, these approaches often face challenges in achieving accelerated or real-time diagnostics, particularly for plasmas that exhibit significant departures from local thermal equilibrium.

In this light, our group is making significant efforts to integrate OES-based diagnostic techniques with data-driven machine learning (ML) methods as a potential solution for accelerated diagnostics. This approach can aid in the development of more efficient nonthermal atmospheric pressure plasma systems [5, 6]. ML offers powerful tools for real-time diagnostics of low-temperature plasmas, capable of extracting meaningful information from complex datasets. By training algorithms on large spectral datasets.

Several recent studies have demonstrated the remarkable versatility of ML in addressing challenges in diagnosing non-thermal plasmas. For instance, Gidon *et al* successfully utilized linear regression to predict the rotational and vibrational temperatures of an atmospheric pressure plasma jet using OES data, showcasing the potential of ML for extracting key plasma parameters from readily available spectral information. In the same study, k-means clustering, an unsupervised ML technique, was employed to distinguish between conductive and insulating substrates using OES data. This capability is valuable for applications where substrate properties significantly influence plasma behavior, enabling more precise control and optimization of plasma processes [7].

Another significant contribution is from Park *et al* [8], who developed an ML-based virtual metrology model for real-time electron temperature and density monitoring during plasma nitridation using in-situ OES data. Their work highlights the potential of combining OES with ML for process monitoring and control in industrial settings, enhancing efficiency and product quality. Additionally, Carter *et al* [9] employed supervised learning methods to predict the accuracy of analyte recovery in inductively coupled plasma optical emission

spectrometry (ICP OES) based on non-analyte signals from the plasma. This study showcases the broader applicability of ML in tackling challenges related to matrix effects and calibration in analytical techniques.

These case studies underscore the feasibility and effectiveness of integrating ML with OES for non-invasive, real-time plasma diagnostics, addressing some of the grand challenges in harnessing the power of low-temperature plasmas. Building on these advancements, our study presents a proof-of-concept application of ML for predicting synthetic spectra from a multi-jet NTAPPJ system. We designed a quadruple plasma jet system to explore scalability, utilizing input parameters such as gas flow rate, voltage, duty cycle, and % air to predict emission peaks of key plasma species.

To enhance the efficiency of our predictive models, we employed principal component analysis (PCA) to reduce the dimensionality of the dataset. This approach facilitated the training of Random Forest (RF) and deep neural network (DNN) models, which successfully predicted the synthetic spectra of important emission lines—including OH(A-X), Ar-I, O-777 nm, and N₂(C-B). Here, the OH(A-X) emission corresponds to the hydroxyl radical (OH) transitions, which are crucial indicators of reactive oxygen species in the plasma. The O-777 nm emission line represents atomic oxygen radicals, essential for various oxidation processes. The N₂(C-B) emission pertains to nitrogen molecules, providing insights into the nitrogen species present in the plasma. These species play significant roles in plasma chemistry and are vital for sterilization, surface modification, and environmental remediation applications [10]. By accurately predicting the emissions of these radicals and molecules, our models contribute to a deeper understanding and control of the plasma processes.

In addition to spectral prediction, we implemented a global optimization strategy using differential evolution, coupled with the trained ML models, to identify the most effective combinations of input parameters that maximize specific emission features, particularly OH(A-X). This inverse modeling framework offers a powerful tool for real-time tuning of non-thermal plasma devices, enabling adaptive control over radical generation. Our work underscores the relevance of ML-based optimization in achieving not only accurate spectral forecasting but also active process control, potentially transforming nonthermal plasma technology into a more intelligent, autonomous, and application-specific platform.

2. Methodology

2.1. Experimental setup and data acquisition

The reactor is a four-jet argon non-thermal atmospheric-pressure plasma jet (multi-Ar-NTAPPJ) array (figure 1). The four jets are mounted in a custom 3D-printed manifold that serves as a common gas plenum and mechanical holder, ensuring uniform distribution of the feed gas to each jet and precise coaxial alignment. Each jet consists of a quartz tube (dielectric barrier) with a stainless-steel ring ground electrode wrapped on the outer surface (ring height: 1 cm; outer

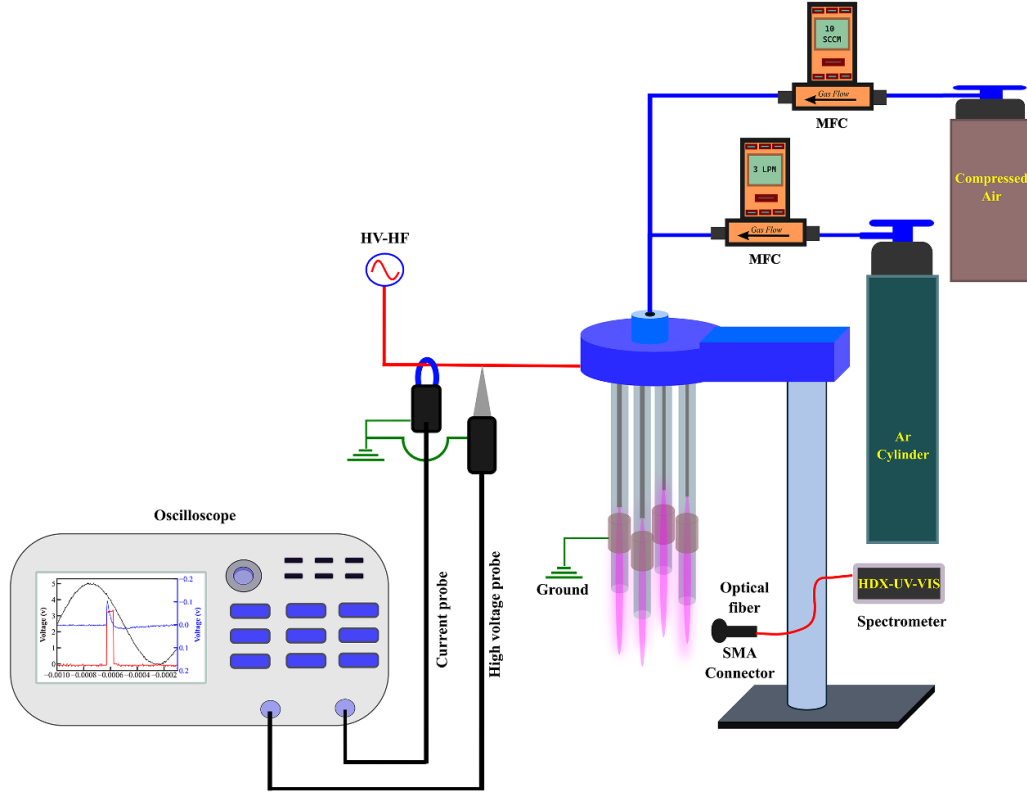


Figure 1. Experimental setup of a Multi-Jet Plasma system equipped with optical and electrical characterization.

diameter: 0.5 cm; inner diameter: 0.31 cm) and a central high-voltage (HV) electrode formed by a hollow stainless-steel blunt needle mounted coaxially inside the tube. The hollow HV needle serves both as the powered electrode and the gas feed: argon (and metered air) enters the 3D-printed manifold from a 6 mm line, flows through the internal lumen of each needle, and exits through the axial orifice at the blunt tip directly into the discharge region. The needle shank is insulated along its length, leaving only the distal tip exposed; all four needles are tied to the same HV output (common node), and all four ring electrodes are bonded to ground. The discharge is driven by a high-frequency HV supply (Plasma Generator G2000, Redline Technologies, Germany) delivering a near-sinusoidal waveform at 215 ± 0.1 kHz. Unless otherwise stated, voltages are reported as *peak-to-peak* V_{pp} measured at the powered electrode. To monitor the operating conditions, the voltage was recorded using a Tektronix HV probe (model: P6015A), while the current was measured with a Tektronix current probe (model: TCP312A). A Tektronix oscilloscope (model: TBS2104B) with a bandwidth of 100 MHz and a sampling rate of 2 GS s^{-1} captured the voltage and current waveforms.

For OES, time-integrated measurements were obtained using an Ocean-HDX-UV-VIS spectrometer (Ocean Optics, Inc. USA) with a spectral resolution of 0.7 nm (FWHM) and a wavelength range from 200 nm to 810 nm. The emitted radiation from the plasma was collected and transmitted to the spectrometer via a $1000 \mu\text{m}$ core-diameter optical fiber,

encased in a custom sleeve to restrict its field of view. A DH-3P-BAL-CAL light source from Ocean Optics was employed for relative intensity calibration. Background noise and spectrometer response corrections were applied to all OES data, with multiple measurements conducted to assess uncertainties in the plasma parameters.

The OES measurements were performed across a broad operating space, including argon flow rates from 3 to 10 LPM, *peak-to-peak* voltages between 3 and 9 kV (V_{pp}), and duty cycles between 17.5% and 35%. Air was introduced at 0–100 sccm to modulate ambient-gas involvement in the discharge, for comparability across different Ar gas flow rates. Each spectrum covered 200–810 nm with a 10 ms integration time, ensuring clear identification of emission lines. Key peaks were extracted from the N_2 second positive system ($C^3\Pi_u \rightarrow B^3\Pi_g$, 330–380 nm), Ar I transitions between $3p^5 4p$ and $3p^5 4s$ (690–810 nm), and O I at 777.5 nm [$2s^2 2p^3(^4S^0) 3p^5 P \rightarrow 2s^2 2p^3(^4S^0) 3s^5 S^0$]. The intensities of these peaks were cataloged by their known electronic transitions and used as supervised-learning labels, with the corresponding experimental inputs (flow rate, voltage, duty cycle, and % air) as features. A representative OES spectrum of the Ar-multijet at 3 LPM is shown in figure 2.

At each operating point, three consecutive spectra were recorded. The variations among repeats were minimal, typically within $\sim 3\%$ – 4% for all considered lines. To minimize label noise and prevent train–test leakage from near-duplicate measurements, the *average* peak intensities were

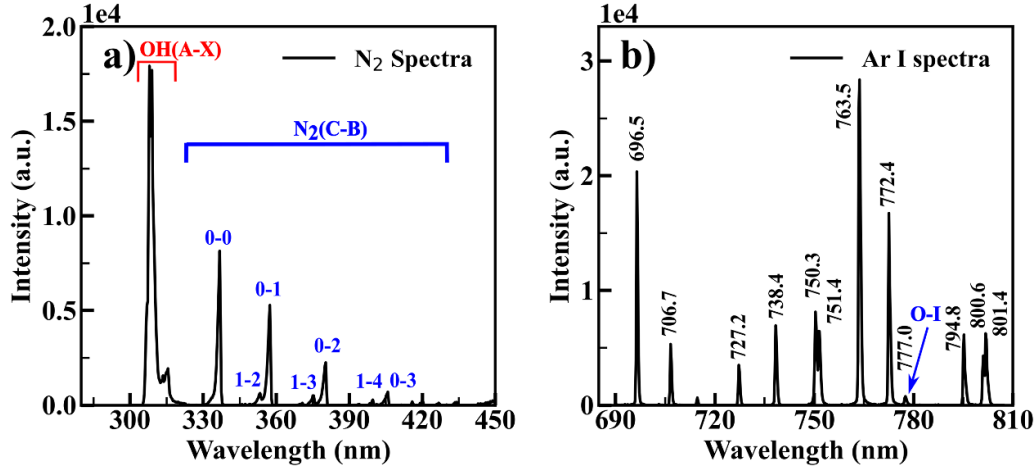


Figure 2. OES measurements of a typical Ar-multijet (a) from OH (A)–(X) and nitrogen (C)–(B) molecule, (b) from Ar neutral atom (Ar-I) excited states. Emission from atomic oxygen is also observed. The same is observed for all the other spectra captured.

used for subsequent analysis and model training. In total, the dataset consists of 1,500 labeled (averaged) measurements, each representing a unique combination of input parameters and corresponding emission intensities, which form the basis for training and evaluating the machine-learning models for OES prediction and simulation under the investigated plasma conditions.

2.2. ML methods

ML techniques are broadly classified into supervised, unsupervised, and semi-supervised methods. Supervised learning involves training models on labeled datasets, where each input is associated with a known output. Techniques such as DNNs and RFs fall into this category, as they learn mappings from input features to target outputs and are widely used for regression, classification, and predictive modeling tasks [11, 12]. Unsupervised learning, on the other hand, is applied to unlabeled data and aims to uncover intrinsic patterns, groupings, or structures within the dataset. A commonly used statistical method in this context is PCA, which, while not a learning algorithm in the strict sense, serves as a powerful tool for dimensionality reduction by projecting high-dimensional data onto a lower-dimensional space that retains the maximum variance [13, 14]. Semi-supervised learning combines elements of both approaches by utilizing a small amount of labeled data alongside a larger pool of unlabeled data to improve learning performance, particularly in scenarios where labeling is expensive or time-consuming [15].

2.2.1. PCA. PCA is a widely used technique for reducing the dimensionality of datasets while preserving as much variability as possible. Dimensionality reduction is essential in many data analysis tasks, as high-dimensional data often suffer from the ‘curse of dimensionality.’ As the number of features increases, data points become more sparse in the feature space, making it challenging to identify meaningful patterns [16]. This sparsity can lead to inefficiencies in ML models, including increased computational costs, decreased

generalization, and diminished performance of distance-based algorithms. Moreover, high-dimensional spaces require larger datasets to prevent overfitting, as models may capture noise instead of underlying relationships [17]. PCA mitigates these challenges by transforming the original high-dimensional data into a lower-dimensional representation while retaining the most significant variance. This process enhances computational efficiency and model interpretability and reveals hidden structures within the data. PCA achieves this by reorienting the data into a new coordinate system defined by principal components (PCs), representing the directions of maximum variance [18].

As illustrated in figure 3, the data in the x_1 – x_2 coordinate system is transformed into the new PCs φ_1 and φ_2 . Here, φ_1 captures the largest variance in the data, and φ_2 , orthogonal to φ_1 , captures the remaining variance [18]. Mathematically, PCA begins by centering the data, which involves subtracting the mean of each feature from the data points. Given a dataset $D \in \mathbb{R}^{N \times 2}$ (where N is the number of samples and 2 represents the two features, x_1 and x_2), the covariance matrix Σ is computed as [19]:

$$\Sigma = \frac{1}{N-1} D^T D, \quad (1)$$

where D is the mean-centered data matrix and D^T represents the transpose of D . The next step is to calculate the eigenvectors and eigenvalues of the covariance matrix Σ . The eigenvectors correspond to the PCs φ_1 and φ_2 , while the eigenvalues represent the variance explained by each component. Mathematically, the eigenvalue decomposition is given by:

$$\Sigma \varphi_i = \lambda_i \varphi_i, \quad (2)$$

where φ_1 and φ_2 are the eigenvectors (PCs), and λ_1 and λ_2 are the eigenvalues corresponding to the amount of variance captured by each component. In figure 3, φ_1 is the direction along which the variance is maximized, while φ_2 is orthogonal to φ_1 and captures the remaining variance. By transforming the data into this new coordinate system, PCA simplifies the

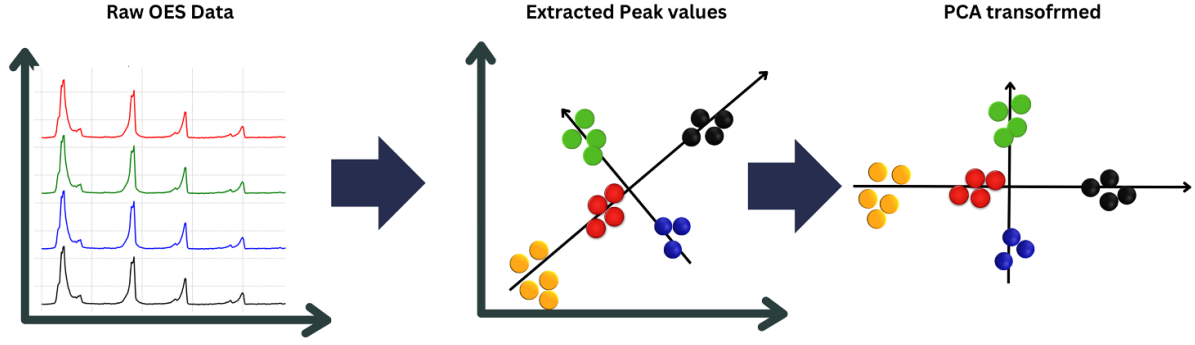


Figure 3. A geometric illustration of PCA in the x_1 - x_2 space, showing the PCs φ_1 and φ_2 , which represent the directions of maximum variance.

dataset while retaining most of its variability [14]. The projection of data points onto the PCs φ_1 and φ_2 results in PC scores, allowing for effective dimensionality reduction. These projections are determined by the PCA loadings, which represent how much each original variable contributes to the PCs. For a given eigenvector φ_i , the loadings are denoted by a_{ij} , and they show how the original features combine to form each PC. If the original features are $x_1, x_2, \dots, x_n$, the first PC PC_1 can be written as [20]:

$$PC_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n, \quad (3)$$

where a_{1j} represents the loading coefficients for the first PC. For dimensionality reduction, we typically retain only the PCs associated with the largest eigenvalues, as these capture the most significant variance in the data. This results in a reduced dataset where each sample is projected onto the new PC space. The transformation from the original data space to the PCA space is given by [18]:

$$p_\varphi = (p_x - \mu_x) \varphi_{PCA}, \quad (4)$$

where p_x is any data point, μ_x is the mean of the original data, and φ_{PCA} is the matrix of selected eigenvectors (PCs). The result is a lower-dimensional representation of the data.

The number of PCs to retain can be determined by analyzing the eigenvalues, which represent the variance captured by each component. A common heuristic is to retain enough components to account for a specified percentage of the total variance, often 95%. Mathematically, the percentage of variance explained by the i th PC is:

$$r_i = 100 \times \frac{\lambda_i}{\sum_{j=1}^n \lambda_j}. \quad (5)$$

In practice, we retain the first m components that explain the desired amount of variance, discarding the rest. This projection leads to significant dimensionality reduction while preserving the most important structures in the data. The PCA loadings provide insights into how each original feature contributes to the reduced representation, which is valuable for interpreting the transformed data.

2.2.2. DNN. A DNN is an artificial neural network that models complex relationships between inputs and outputs by passing data through multiple layers of interconnected neurons [21]. At the heart of a DNN is the neuron, which takes in inputs, applies weights and biases, and produces an output through a non-linear activation function. This non-linearity is crucial as it allows the network to learn and represent complex, non-linear relationships in the data.

The schematic structure of the DNN used in the present study is given in figure 4. The input to the network is a vector $\mathbf{x} \in \mathbb{R}^{N_0}$, where N_0 is the number of input features. Each layer consists of several neurons, and each neuron in layer i computes an output using the following formula:

$$\mathbf{z}^{(i)} = \sigma \left(\mathbf{W}^{(i)} \mathbf{z}^{(i-1)} + \mathbf{b}^{(i)} \right), \quad (6)$$

where $\mathbf{W}^{(i)} \in \mathbb{R}^{N_i \times N_{i-1}}$ is the weight matrix for layer i , $\mathbf{b}^{(i)} \in \mathbb{R}^{N_i}$ is the bias vector, N_i is the number of neurons in layer i , and σ is the activation function. Common activation functions include ReLU (Rectified Linear Unit) and sigmoid. The vector $\mathbf{z}^{(i-1)}$ represents the output from the previous layer.

The present work utilizes a DNN model with five hidden layers, along with input and output layers, as shown in figure 4. The input layer consists of four nodes, each representing one of the input features: flow rate, voltage, % air, and duty cycle. The output layer has four nodes, each corresponding to one of the first four PCs. The first three hidden layers, L_2, L_3 , and L_4 , consist of 1024 neurons each, while the last two hidden layers, L_5 and L_6 , contain 512 neurons each.

The process of passing the input through each layer in this manner is called forward propagation. In the final layer, the output $\mathbf{y}$ is produced, which represents the model's prediction [22]. Initially, the network's weights and biases are randomly initialized, resulting in inaccurate predictions. The goal of the training process is to adjust these weights and biases to minimize the error between the predicted output $\mathbf{y}$ and the actual target values.

To reduce this error, we used a technique called backward propagation. Backward propagation computes the gradients of the loss function with respect to each weight and bias in the network, which tells us how much each parameter needs to change to reduce the loss [23]. These gradients are calculated using the chain rule of calculus, which propagates the error

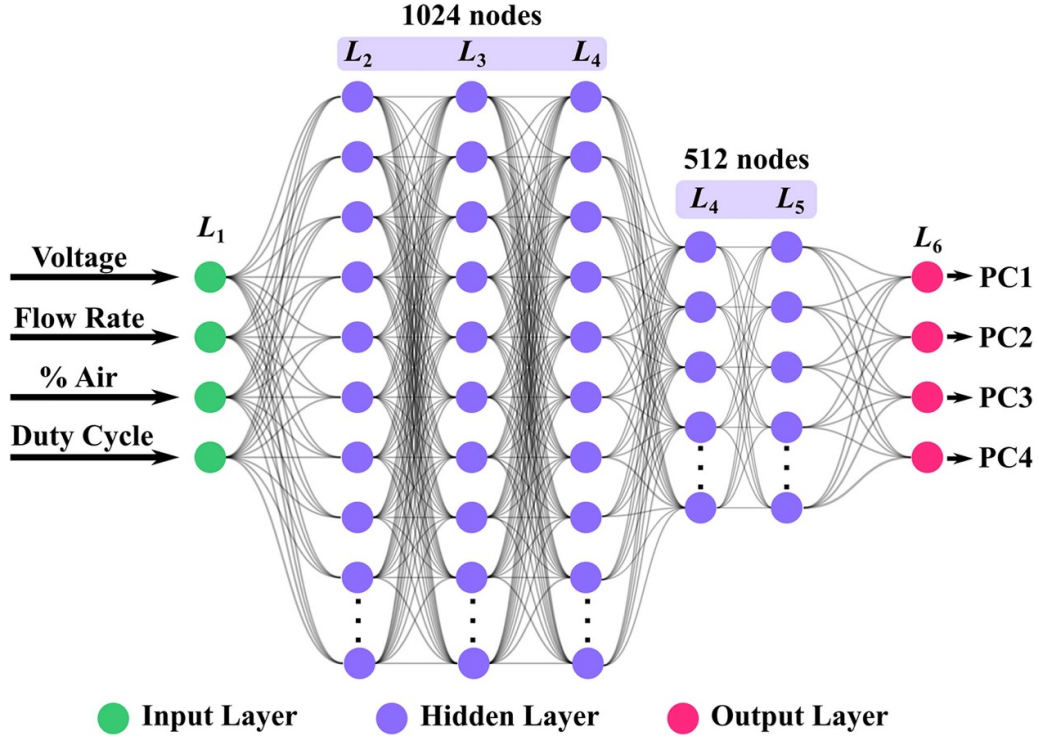


Figure 4. Illustration of a feed-forward neural network with four input nodes, five hidden layers, and four output nodes. first three hidden layer consists of 1024 neurons, where each node applies an activation function to an affine transformation of the inputs from the previous layer, and the last two hidden layers consists of 512 neurons. The network propagates information sequentially through the layers, and the number of hidden layers defines the depth of the network. The DNN predicts the first four principle components as outputs.

backward through the network, starting from the output layer and moving toward the input layer. For each weight $\mathbf{W}^{(i)}$, the gradient is computed as:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(i)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}^{(i)}} \frac{\partial \mathbf{z}^{(i)}}{\partial \mathbf{W}^{(i)}}, \quad (7)$$

where $\mathcal{L}$ is the loss function. Similarly, gradients for the biases are computed. Once the gradients are obtained, they are used to update the weights and biases in a direction that reduces the loss [22]. This is done using a method called gradient descent, where each parameter is updated according to the rule:

$$\mathbf{W}^{(i)} \leftarrow \mathbf{W}^{(i)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(i)}}, \quad (8)$$

where η is the learning rate, a hyperparameter that controls how large the updates are during each step of gradient descent. If the learning rate is too high, the network might overshoot the optimal solution, and if it is too low, training may take an excessively long time.

A more advanced optimization algorithm commonly used to improve the efficiency of training is Adam (Adaptive Moment Estimation). Adam improves upon basic gradient descent by adjusting the learning rate for each parameter based on the first moment (mean) and second moment (uncentered variance) of the gradients. This makes Adam particularly effective for training deep networks, as it helps avoid issues

like slow convergence or large oscillations [24]. The update rule for Adam is:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}, \quad (9)$$

where $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected estimates of the first and second moments, respectively, and ϵ is a small constant to prevent division by zero.

By continuously performing forward propagation to make predictions, computing the loss, and using backward propagation with Adam or another optimization algorithm to adjust the weights and biases, the network gradually improves its performance. This iterative process allows the DNN to learn from the data, minimizing the error over time and producing more accurate predictions as the network converges toward an optimal set of parameters.

In summary, a DNN takes input data, processes it through neurons in successive layers using forward propagation, computes the loss, and adjusts its parameters using backward propagation. The learning rate and optimization algorithm, such as Adam, are key to ensuring efficient and effective training, enabling the network to learn complex patterns from the data.

2.2.3. RF RF model is a type of ensemble learning method, where multiple decision trees are constructed and aggregated to produce a more robust and accurate prediction. The

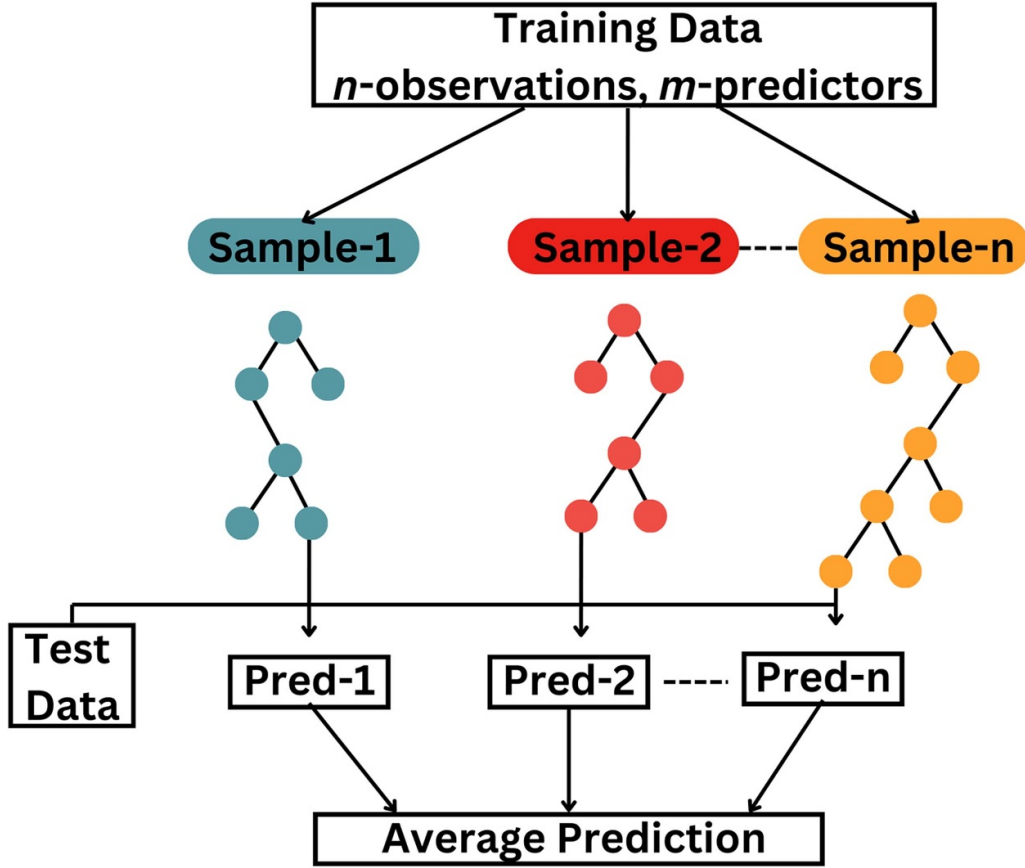


Figure 5. Conceptual representation of a regression decision tree.

‘random’ aspect of RF comes from two sources: random sampling of training data for each tree (bagging) and random selection of features at each node [25]. A decision tree, the base model of the RF, works by making a series of decisions to split the data into smaller subsets at each step as shown in figure 5. Each split is based on a condition that maximizes information gain (in classification tasks) or minimizes the mean squared error (MSE) (in regression tasks). Starting from a root node (which contains all the data), the decision tree divides the dataset into two branches called daughter nodes. The process continues until the model reaches leaf nodes, which represent the final prediction [26].

For a dataset Q_m consisting of N samples (x_i, y_i) , where x_i is a vector of features and y_i is the target value, the MSE for any subset Q of data is calculated as [27, 28]:

$$H(Q) = \frac{1}{N_Q} \sum_{i \in Q} (y_i - \bar{y}_Q)^2, \quad (10)$$

where N_Q is the number of samples in Q , and $\bar{y}_Q$ is the mean of the target values in Q :

$$\bar{y}_Q = \frac{1}{N_Q} \sum_{i \in Q} y_i. \quad (11)$$

At each decision step, the algorithm evaluates all possible splits of a feature to minimize the MSE. For a feature j and a split point t , the dataset Q_m is divided into two subsets:

$$Q_{\text{left}}(\theta) = \{(x_i, y_i) \in Q_m \mid x_{ij} \leq t\} \quad (12)$$

$$Q_{\text{right}}(\theta) = Q_m \setminus Q_{\text{left}}(\theta) \quad (13)$$

where θ represents the splitting condition (j, t) . For these subsets, the MSE is recalculated, and the split that minimizes the weighted MSE of the two branches is chosen. The weighted MSE is computed as:

$$G(Q_m, \theta) = \frac{N_{Q_{\text{left}}(\theta)}}{N_{Q_m}} H(Q_{\text{left}}(\theta)) + \frac{N_{Q_{\text{right}}(\theta)}}{N_{Q_m}} H(Q_{\text{right}}(\theta)). \quad (14)$$

The algorithm chooses the split $\theta^* = (j, t)$ that minimizes $G(Q_m, \theta)$, leading to the next branching point. This process is recursively applied until the model reaches a stopping criterion, such as a maximum tree depth or a minimum number of samples per node, resulting in a fully developed decision tree [29].

In an RF, multiple decision trees are trained using different subsets of the training data and features. This introduces randomness into the model, helping to reduce overfitting and

variance. Each tree in the forest is constructed using a technique called bagging (Bootstrap Aggregating), where a random sample of the original dataset is selected (with replacement), and a subset of features is chosen randomly at each node to determine the best split. This creates multiple diverse decision trees and reduces the correlation between trees, which is crucial for the performance of RFs [30].

Once the individual trees are trained, the RF model combines their predictions by averaging them in the case of regression tasks:

$$y = \frac{1}{c} \sum_{k=1}^c y_k, \quad (15)$$

where c is the number of decision trees, and y_k is the prediction from the k^{th} tree for a given input instance. This aggregation reduces the variance of the model, as the average of multiple models is less sensitive to noise and outliers than a single model.

The effectiveness of the RF is influenced by several hyperparameters, such as `n_estimators` (the number of decision trees in the forest), `max_features` (the number of features considered for splitting at each node), `max_depth` (the maximum depth of each decision tree), and `min_samples_split` and `min_samples_leaf` (which control how many samples are required to split a node or to form a leaf). These hyperparameters allow fine-tuning of the model's complexity and performance.

RF models are particularly useful because they can handle both classification and regression tasks and are resistant to overfitting due to their inherent randomness. Moreover, RF models are robust against noise and can deal with high-dimensional data, making them a versatile tool in ML. They also provide a measure of feature importance and offer out-of-bag (OOB) error estimation, a form of built-in cross-validation, which can be valuable for model evaluation.

2.3. Model evaluation: performance metrics and visualization analysis

To evaluate the effectiveness of the RF and DNN models, we employ a set of metrics and visualizations to gauge performance. Two primary metrics, MSE and the Coefficient of Determination (R^2), are used to quantify model accuracy. The MSE, measuring the average squared difference between observed and predicted values, is calculated as follows:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_{\text{observed},i} - y_{\text{predicted},i})^2. \quad (16)$$

The R^2 value, which ranges from 0 to 1, represents the proportion of variance in the observed data explained by the model. An R^2 of 1 indicates a perfect fit between observed and predicted values. It is computed as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_{\text{observed},i} - y_{\text{predicted},i})^2}{\sum_{i=1}^N (y_{\text{observed},i} - \bar{y}_{\text{observed}})^2}. \quad (17)$$

In these expressions, $y_{\text{observed},i}$ and $y_{\text{predicted},i}$ denote the observed and predicted values at the i^{th} instance, while $\bar{y}_{\text{observed}}$ represents the mean of the observed values. Here, N is the total number of data points.

These metrics offer a quantitative perspective on the models' learning effectiveness from the training data. To complement these metrics, visual tools such as residual plots and scatter plots of predicted vs. true values are utilized. Residual plots help in identifying any patterns in the errors, while scatter plots provide a visual correlation between actual and predicted outcomes. Together, these visualizations provide a comprehensive assessment of the model's performance, highlighting its predictive accuracy and areas that may require refinement.

3. Results and discussion

3.1. PCA analysis of acquired OES

The use of PCA in this study highlights its utility in reducing the dimensionality of complex plasma spectral data. By transforming the correlated spectral peaks into uncorrelated PCs, PCA captures the maximum variance, simplifying the dataset while preserving critical information [18, 31]. This approach revealed hidden patterns, such as the interaction between nitrogen and argon emissions, without relying on labeled data. The first PC (PC1) accounted for 88% of the total variance, while the second (PC2) added 8.6%, together explaining approximately 96.6% of the dataset's variability. This rapid accumulation of explained variance indicates that the first two PCs were sufficient for capturing the majority of the dataset's complexity, reducing the need for additional components. Here we have decided to choose the First 4 PCs, which is capturing almost 99% variance in the data. The rationale behind this is capturing all the variance is essential for accurate prediction, especially when data sets are small. This dimensionality reduction was essential for efficient ML model development while retaining the most important features of the spectral data [32].

The PCA data visualization presented in figure 6 focuses on the relationship between PC1 and PC2, as these two PCs capture over 95% of the total variance in the dataset. The scatter plot in figure 6(a) visually represents how these two PCs relate to each other, with data points color-coded according to the flow rate. The clear separation of points along the PC1 axis indicates that PC1 accounts for a significant portion of the variance, as reflected in the distinct clustering of points. This suggests that the behavior of the plasma, as captured by PC1, is strongly influenced by the flow rate. Additionally, PC2 captures a secondary, but still important, variation in the dataset, providing further insight into the plasma dynamics under varying operational conditions.

To understand the significance of these components, we first examine the correlation matrix shown in figure 7, which provide a visual representation of the relationships between operating parameters, PCs, and key spectral peaks. On the left, we observe the correlation between the operating parameters (flow rate, voltage, duty cycle, and % Air) and the first four PCs (PC1 to PC4). The right side shows the correlation between the operating parameters and selected plasma emission peaks,

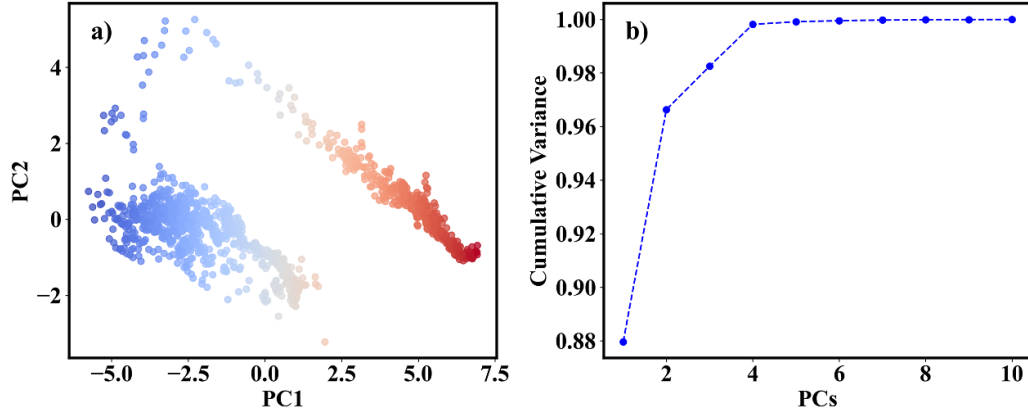


Figure 6. (a) Scatter plots of the first two principal components (PCs) derived from PCA of the standardized plasma emission peak values (b) cumulative variance explained by the principal components. The first two components account for approximately 96% of the total variance.

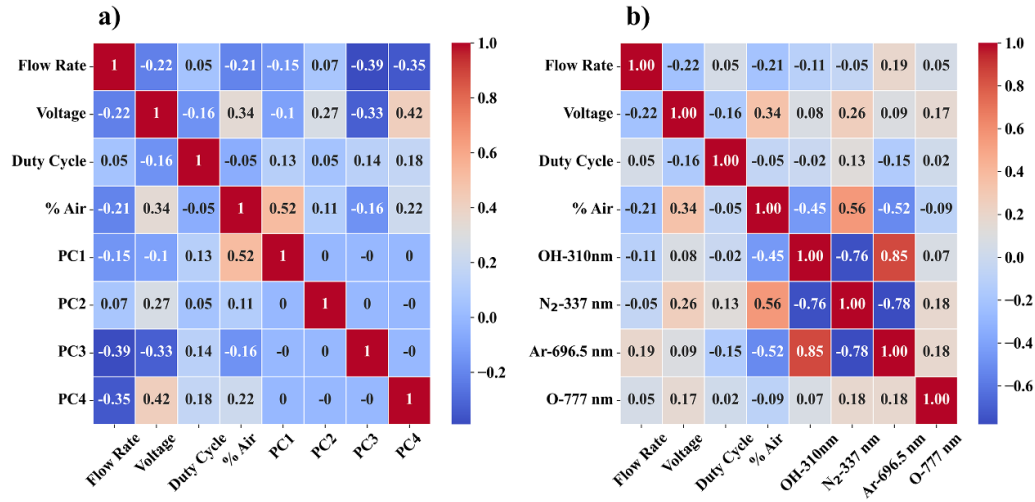


Figure 7. Heatmap illustrating the correlation between operating parameters and principal components. (a) Correlation between operating parameters (Flow Rate, Voltage, Duty Cycle, and % Air) with principal components (PC1 to PC4) (b) correlation between operating parameters and plasma emission peaks, and key spectral peaks (OH, N₂, Ar, and O emissions).

including OH(A-X), N₂(C-B), Ar-I 696.5 nm, and O-I 777 nm. Mathematically, the correlations between the variables and features can be expressed as the correlation coefficient (r), calculated using the following equation:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}} \quad (18)$$

where X_i and Y_i represent individual data points for the two variables being correlated, and $\bar{X}$ and $\bar{Y}$ are their respective means. This equation provides a measure of the linear relationship between the variables, helping to understand which experimental conditions most strongly influence the plasma behavior as captured by the PCs.

The PCA combined with correlation heatmaps provides a comprehensive understanding of how various operating parameters and plasma emission peaks contribute to the overall plasma behavior. As depicted in figures 7(a) and (b), the heatmaps highlight significant correlations between the operating parameters, the PCs, and the selected emission peaks.

In the first heatmap, % air mixture shows a strong positive correlation (0.52) with PC1, indicating that % air significantly influences the variance captured by this PC. This implies that increasing % air in the plasma has a substantial effect on the primary variance within the dataset. Voltage exhibits a notable correlation with PC2 (0.27), suggesting its impact on the plasma behavior captured by this component. PC3 and PC4, which account for less variance, demonstrate more complex and weaker correlations with the operating parameters. This suggests that they might capture more subtle plasma dynamics, potentially including noise or minor experimental variations. The second heatmap shows the relationships between the same operating parameters and selected emission peaks. Notably, % air mixture has a strong negative correlation (−0.45) with the OH(A-X) peak and a positive correlation (0.56) with N₂(C-B). This correlation pattern indicates that as the % air content increases, the emission intensity of OH decreases while nitrogen molecular emissions become more dominant. This aligns with the physical chemistry of the plasma, where increased oxygen or air in the plasma reduces hydroxyl emissions while promoting nitrogen excitation [33].

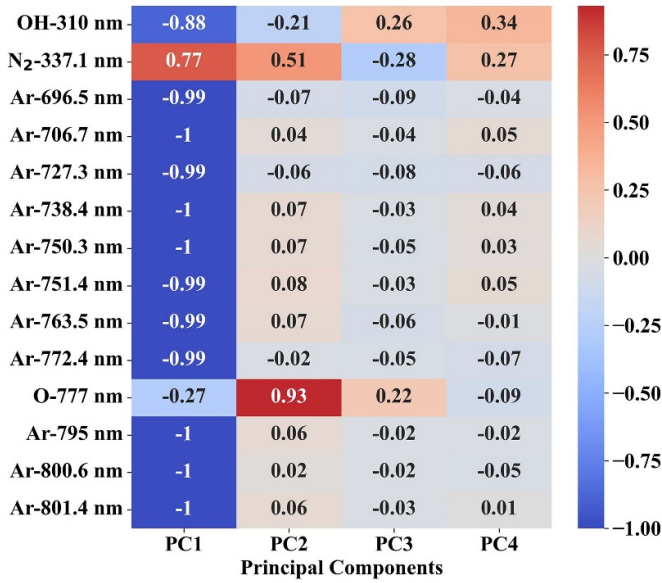


Figure 8. PCA loadings heatmap showing the contribution of key plasma emission peaks to the first four principal components (PC1 to PC4). Negative loadings (in blue) indicate an inverse relationship between the spectral peaks and the principal component, while positive loadings (in red) indicate a direct relationship.

Additionally, the argon emission line at 696.5 nm displays a strong positive correlation with OH(A-X), indicating that when argon emissions are intense, OH emissions are also high. This suggests the importance of argon in sustaining plasma species like OH, which are key players in many AOP-based plasma applications. In addition, the PCA loadings heatmap, shown in figure 8, further enhances the understanding of how specific emission peaks contribute to the variance explained by PC1 and PC2. The loadings, mathematically represented as:

$$\text{Loading}_{ij} = \frac{\text{eigenvector}_{ij}}{\sqrt{\lambda_i}} \quad (19)$$

where eigenvector_{ij} is the element of the eigenvector corresponding to the j th variable and the i th PC, and λ_i is the eigenvalue for that PC. These loadings quantify how much each original variable contributes to each PC.

From figure 8, the PCA loadings reveal two orthogonal, physics-meaningful modes. PC1, which explains the most variance, carries large negative loadings for nearly all Ar lines (e.g. 696.5, 750.4, 763.5 nm) and for OH(A-X, 309–310 nm), while N₂(C-B, 337 nm) loads positively. Moving in the +vePC1 direction therefore corresponds to increasing in % air (N₂/O₂) fraction: N₂ density rises and enables Ar* → N₂(C) energy transfer, increasing the 337 nm band, whereas N₂/O₂ dilute Ar and efficiently quench Ar(4p) states and OH(A), reducing their emissions [34]. PC2 is dominated by the O I 777 nm triplet (loading ≈ 0.93), with a secondary positive loading for N₂(C-B) and a weak negative loading for OH. This component reflects the voltage lever: higher the voltage, the more the field, enhancing O₂ dissociation and O-atom excitation that feed the 777 nm line, modestly elevating N₂(C-B) while conditions that maximize O 777 nm can

slightly suppress OH(A) via increased quenching and local heating. (PC signs are arbitrary; interpretation relies on relative signs.) By identifying the key drivers of variance in the dataset, this analysis informs the selection of the most relevant features, enabling DNNs and RF models to predict plasma behavior accurately under varying conditions. The combination of PCA and correlation analysis offers a foundation for understanding how molecular and atomic species interact within the plasma and lays the groundwork for more advanced predictive models.

3.2. RF for predicting PCA components

The application of RF regression to predict the PCs derived from plasma spectral peaks is an essential step in reducing the dimensionality of the dataset and building a robust predictive model. Initially, the dataset comprised 14 plasma emission peaks, which were reduced to four PCs using PCA. This reduction simplified the regression task by capturing over 90% of the variance in the dataset while retaining the most important features. The operating parameters—flow rate, voltage, duty cycle, and % Air—served as the input features for the RF model, while the reduced PCA vectors (PC1 to PC4) were used as the target outputs. The RF model was built using the *RandomForestRegressor* as the base estimator, and the final hyperparameters (table 1) were selected via a randomized search using scikit-learn's *RandomizedSearchCV* over pre-defined ranges with k -fold cross-validation on the training set; the best configuration was then refit on the full training split before evaluation on the 20% test set. A *MultiOutputRegressor* wrapper was employed to handle the multi-target regression task so that the same tuned RF hyperparameters were applied consistently across all four PCs. The RF was trained on 80% of the data, and after training, it was used to predict the PC values for the remaining 20% test set.

Figure 9 shows the True vs. Predicted plots for PC1 to PC4, where the blue points represent the training data, and the red points represent the test data. The diagonal dashed line in each subplot represents the ideal case where the predicted values perfectly match the true values. The proximity of the points to the line indicates the model's performance. In figure 9(a), PC1, which explains the majority of the variance, shows an excellent fit between the predicted and actual values for both the training and testing data. This trend continues for PC2, PC3, and PC4, although there is a slight increase in variance in the residuals for the later PCs, as these components capture more complex and subtle relationships within the dataset.

Table 1. Hyperparameters used in the RF model.

Hyperparameter	Value
Number of trees (n_estimators)	700
Maximum depth (max_depth)	20
Minimum samples to split (min_samples_split)	4
Minimum samples per leaf (min_samples_leaf)	2
Maximum features (max_features)	sqrt
Random state	42

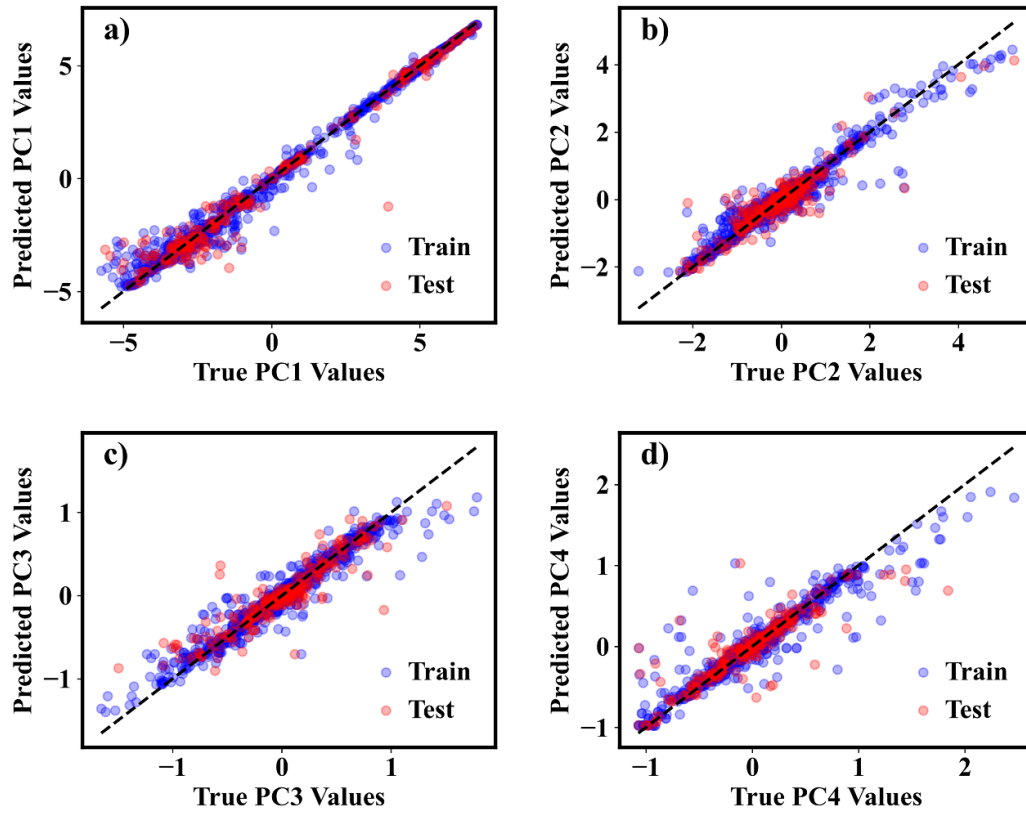


Figure 9. True vs Predicted values for (a) PC1, (b) PC2, (c) PC3, and (d) PC4. The dashed line represents the ideal fit, where True values equals Predicted.

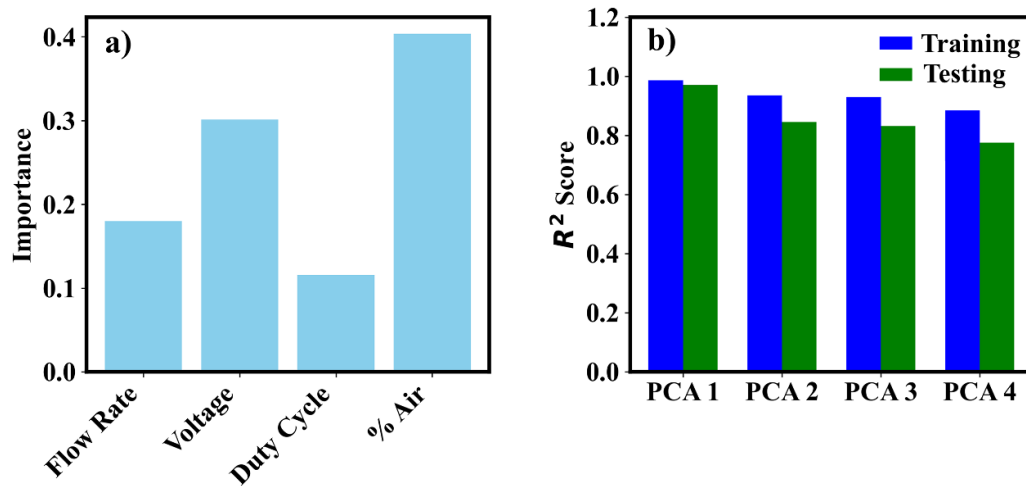


Figure 10. (a) Feature importance plot showing the contribution of each operating parameter to the RF model. % Air and voltage have the highest importance; (b) R^2 scores for the RF model for the training (blue) and test (green) datasets for PC1 to PC4.

To further analyze the RF model, we extracted feature importances alongside the R^2 values as shown in the figure 10. The importance of % Air as the dominant driver of the model's predictions, followed by voltage, with Flow rate and Duty cycle contributing less. This hierarchy is consistent with the PCA loadings. PC1 accounting for the largest share of variance—acts as a composition/air-dilution mode in which N_2 (C-B, 337 nm) loads positively while nearly all Ar lines and OH(A-X, 309–310 nm) load negatively. Increasing the

% air (N_2/O_2) fraction therefore coherently suppresses Ar and OH emission via dilution and collisional quenching (including reduced Ar(2p) populations and efficient quenching of OH(A)) while enabling stronger N_2 excitation directly by electrons and through $Ar^* \rightarrow N_2(C)$ energy transfer [34]. PC2 is dominated by the O I 777 nm triplet (loading ≈ 0.93), reflecting an voltage-controlled pathway: higher voltage increases the field, enhancing O_2 dissociation and O-atom excitation that feed the 777 nm line; N_2 (C-B) shows a secondary positive

projection, whereas OH is slightly negative, consistent with increased quenching and local heating at higher E/N [35]. Accordingly, the RF assigns greatest importance to % air because composition sets the collisional and quenching landscape and thus drives a system-wide, coherent response captured by PC1, while Voltage ranks next as it tunes electron energy and dissociation pathways most visibly along PC2 (via O77 nm).

The R^2 values indicate how well the RF model fits the data. For PC1, which captures the most variance, the R^2 score is close to 1 for both the training and test sets, demonstrating that the RF model performs exceptionally well in predicting this component. The R^2 scores for PC2, PC3, and PC4 are slightly lower, but still show strong predictive performance, with the model generalizing well to the test set. This consistent performance across multiple PCs suggests that the RF model effectively captures the relationships between the operating parameters and the spectral peaks.

The RF model successfully learned the complex relationships between the operating parameters and the plasma spectral peaks through the reduced PCA components. The model's performance, indicated by high R^2 values and well-distributed residuals, confirms its robustness and suitability for predicting plasma behavior. This predictive capability is valuable for real-time diagnostics and optimization of plasma systems, where precise control of operating conditions is crucial.

3.3. DNN for predicting PCA components

In addition to the RF model, a DNN was employed to predict the PCA results based on the plasma operating parameters. The motivation for using the DNN model arises from its ability to capture complex, non-linear relationships between the input features and the output target variables, which in this case are the reduced PCA components representing the spectral peak information.

The architecture of the DNN model consisted of five dense layers with 1024, 1024, 1024, 512, and 512 neurons, respectively, using ReLU activation functions to introduce non-linearities in the learning process. The model was designed to map the four input parameters—flow rate, voltage, duty cycle, and % air—to the four PCA components. The Adam optimizer was used for training, with a learning rate of 0.001, and the loss function was the MSE, which is suitable for regression tasks, and the list of hyperparameters is given in table 2 and the hyperparameters except for number of layers and number of neurons in each layer the parameters were selected via a randomized search using scikit-learn's *RandomizedSearchCV*.

As depicted in figure 11, the training loss steadily decreases over time as the network learns from the data, with validation loss following a similar trend, though with more fluctuations. The training and validation curves suggest that the model converges well, although some overfitting is likely, as seen in the noisier validation loss. The model was trained for 5000 epochs to ensure the network learned the complex patterns in the data.

The performance of the DNN model was evaluated using the coefficient of determination (R^2) for both the training and

Table 2. Hyperparameters used in the DNN model.

Hyperparameter	Value
Number of layers	5 hidden layers
Number of neurons per layer	1024 (3 layers) and 512 (2 layers)
Activation function	ReLU
Optimizer	Adam
Learning rate	0.001
Loss function	MSE
Batch size	64
Number of epochs	5000

testing datasets. As shown in figure 11(b), the R^2 scores for PC1 and PC2 exceed 0.9, demonstrating the model's strong predictive ability for these components, which capture the majority of the variance in the spectral peaks. While lower, the scores for PC3 and PC4 still indicate decent predictive power, highlighting the model's capacity to generalize across the different components.

The scatter plots in figure 12 further illustrate the predictive power of the DNN model. The true PCA values are plotted against the predicted values for both the training and testing datasets in these plots. The diagonal dashed line represents the ideal case where the expected values match the true values. For PC1 (figure 12(a)) and PC2 (figure 12(b)), the points closely cluster around the diagonal line, suggesting a high degree of accuracy in the model's predictions. PC3 and PC4, while showing slightly more dispersion, still exhibit a good overall trend, which highlights the model's ability to generalize even for components that capture less variance.

The DNN model performs very well in predicting the first PCs (PC1 and PC2), accounting for more than 90% of the original dataset's variance. This result demonstrates the utility of DNNs in handling highly non-linear datasets, where traditional linear models like PCA alone may fall short in capturing complex patterns. Though slightly less accurate, the prediction for PC3 and PC4 still provides valuable insights into the system behavior.

The DNN model, combined with PCA for dimensionality reduction, successfully captured the underlying plasma behavior. By leveraging the reduced PCA components, the model was able to learn key relationships between the input operating parameters and the plasma spectral peaks, leading to strong predictive performance.

3.4. Comparison of DNN and RF for simulating OES

The RF and DNN models were implemented to predict the PCs of OES data obtained from a multi-plasma jet system. The predictive capability of these models was assessed based on their performance in simulating the original spectra.

The RF model showed remarkable predictive accuracy across all PCs. It required only 5.6 s to train and exhibited high generalization capability for both train and test data. For instance, in PC1, RF achieved a train MSE of 0.1642 and an R^2 of 0.9866, while for the test set, it gave an MSE of 0.3710

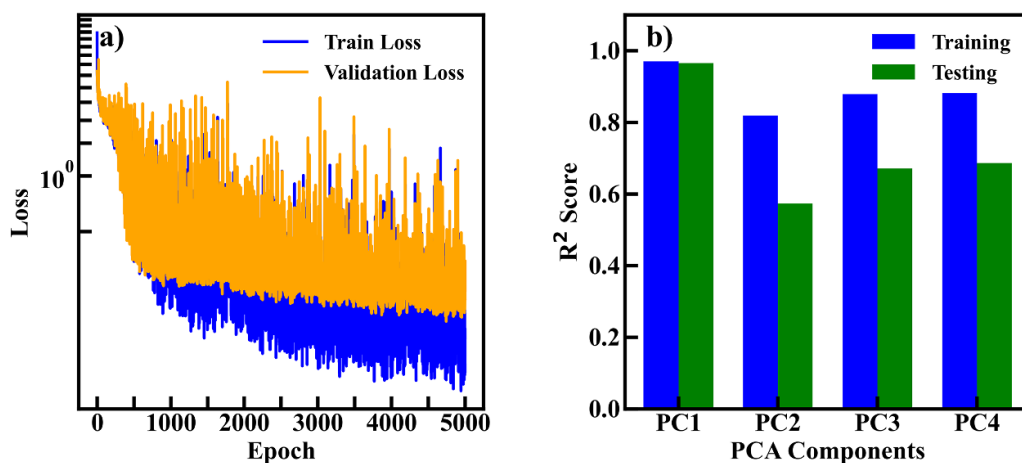


Figure 11. (a) Training and validation loss curves for the DNN model. The model shows a good convergence for both the training and validation data. (b) R^2 scores for the PCA components on the training and testing datasets, indicating strong predictive performance, particularly for PC1 and PC2.

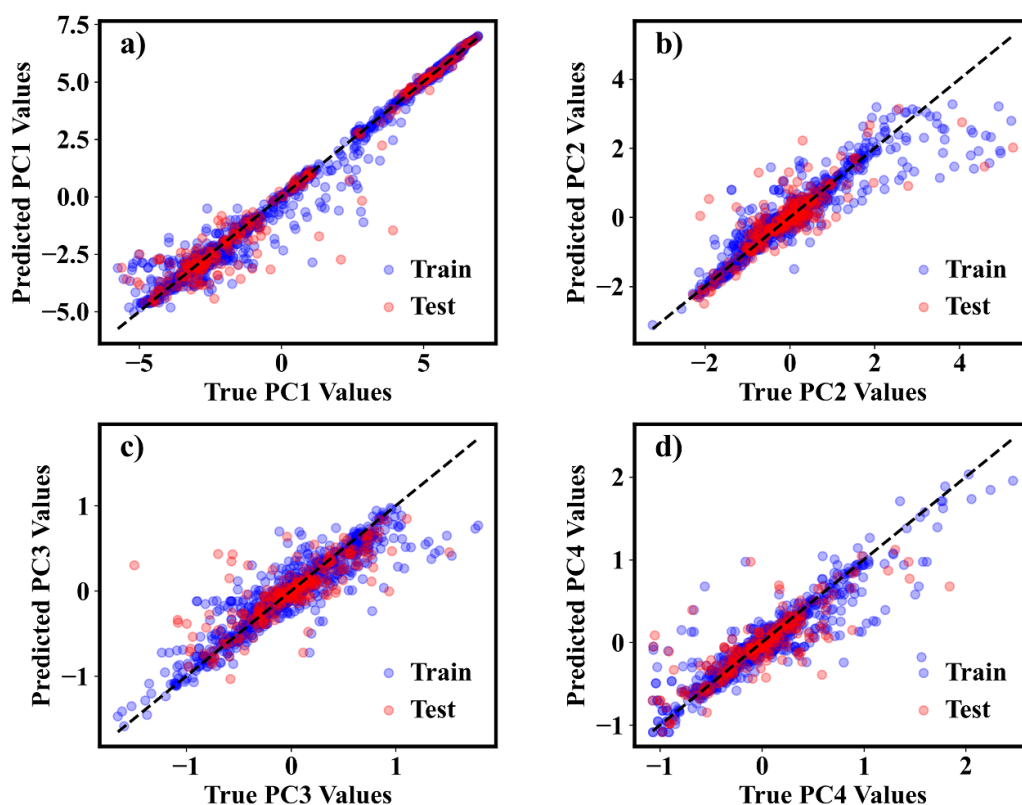


Figure 12. True versus predicted values for the DNN model across the first four PCs: (a) PC1, (b) PC2, (c) PC3, and (d) PC4. The blue points correspond to training data, while the red points correspond to testing data. The dashed black line represents the ideal $y = x$ line, indicating perfect predictions.

and R^2 of 0.9704. Similar performance was observed for other PCs, with RF accurately capturing patterns and generalizing well to unseen data. This is evident in comparing the actual and predicted spectra, as shown in figure 13.

On the other hand, the DNN model took approximately 10 min to train and required a significantly larger amount of data to learn the intricate non-linear relationships present in the OES data. While DNN also performed well on the training data, its test performance slightly lagged behind RF in terms

of generalization. For example, in PC1, the DNN achieved an R^2 score of 0.924 for the test data, which is lower than the RF model. The training and validation loss curves in figure 11 further reveal the fluctuations in the DNN's performance during training, indicating that DNN models require longer training time and larger datasets for reliable predictions.

The comparative analysis between the actual and predicted spectra for both models (figure 13) demonstrates that RF, despite its simpler structure, can achieve predictions that are on

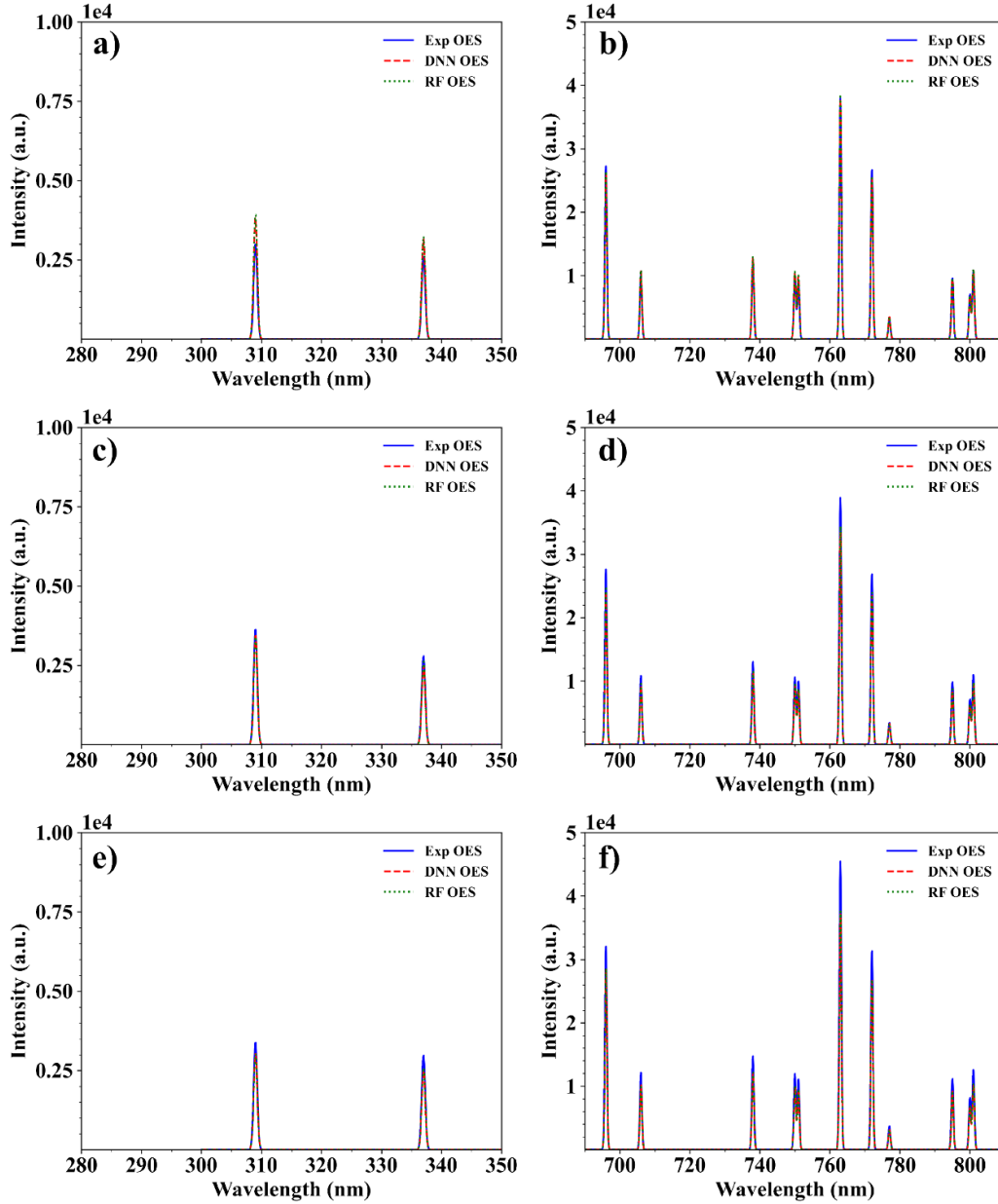


Figure 13. Comparison between experimentally acquired and predicted OES spectra using DNN and RF models. Both models accurately replicate the original spectra across three randomly selected cases from the test dataset, showcasing high fidelity in simulating the experimental data.

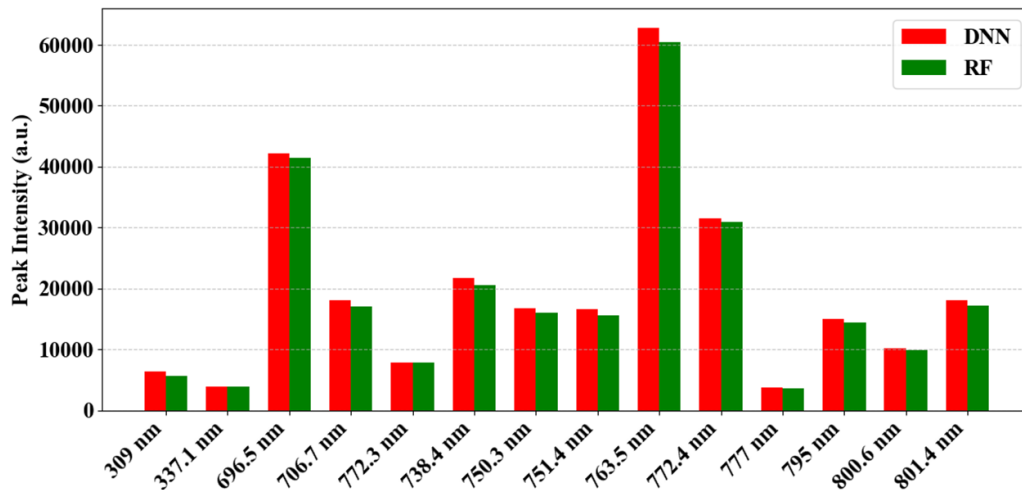
par with or even exceed those of DNN in this particular case. The results highlight the suitability of RF for capturing patterns in relatively smaller datasets and generalizing to unseen conditions with less computational effort.

To contextualize the RF–DNN comparison on our OES dataset, we note that the leading spectral structure is concentrated in the first two PCs ($PC1+PC2 \approx 98\%$ of variance). On these dominant modes, both models achieve high accuracy ($R^2 > 0.98$; figures 10(b) and 11(b) on train data), indicating that the learned input–output mappings are broadly consistent. The small gap that opens for the DNN on $PC2$ – $PC4$ arises from a classical capacity–data trade-off: our feed-forward DNN (trained for a fixed number of

epochs without explicit regularization such as early stopping, dropout, or weight decay) has many more free parameters than effective constraints at $N \approx 1,500$, which increases estimator variance and yields the mild validation-loss fluctuations visible in figure 11(a). RFs, by contrast, embed strong regularization by construction—bagging and feature subsampling reduce variance, and axis-aligned partitions efficiently capture low-order, piecewise interactions among the experimental factors (composition, voltage, flow) common in tabular settings. We emphasize that this result does not generalize against DNNs in principle: neural models typically benefit from larger, more diverse datasets and from domain priors (e.g. physics-motivated features or constraints),

Table 3. Optimized input parameters obtained from DNN and RF models for maximizing OH(A-X) peak intensity in argon plasma jet.

Input parameter	DNN optimal	RF optimal	Difference
Flow rate (slm)	5.4	5.0	0.4
Voltage (kV)	6.0	6.30	−0.31
Duty cycle (%)	34.0	30.0	5.1
% Air (%)	0.0	0.039	−0.039

**Figure 14.** Comparison of predicted emission peak intensities from DNN and RF models using their respective optimized input parameters. The OH(A)–(X) peak was the primary optimization target.

at which point their representation capacity can become advantageous.

3.5. Optimization of input parameters for OH(A-X) enhancement

To investigate the predictive capabilities of the developed regression models and to determine the optimal input settings for maximizing OH(A-X) emission in a nonthermal argon plasma jet, we employed a differential evolution optimization scheme [36]. This was applied independently to both the DNN and RF models, which were previously trained to estimate PC representations of experimentally acquired optical emission spectra. The primary optimization objective was to identify the set of four optimized parameters—gas flow rate, voltage, duty cycle, and % air mixture—that would lead to the highest possible intensity in the OH(A-X) band, following the inverse transformation of the PCA-reduced outputs.

The resulting optimal inputs obtained from both models are summarized in table 3. While both models identified distinct parameter sets, notable differences were observed. The DNN favored a higher flow rate and duty cycle with no air admixture, whereas the RF model optimized towards a higher voltage and a small % air, possibly attempting to exploit locally optimal splits in the training data. The predicted peak intensities obtained using these parameter sets are shown in figure 14. Interestingly, the RF model produced a higher OH(A-X) intensity than the DNN model in this instance, which can be attributed to its ability to perform well

under moderate dataset sizes. Given that the training dataset comprised approximately 1500 samples, the RF model, being non-parametric and inherently robust to overfitting, was more effective at capturing empirical patterns from the available data.

However, the DNN framework remains promising due to its capacity to learn complex nonlinear mappings and generalize across coupled parameter spaces, especially when trained on larger datasets. Although the DNN's predictive performance was slightly lower in this case, its architecture is inherently more adaptable for integration into real-time, self-correcting systems. In such a setup, the model could be periodically updated with incoming spectral feedback from the experiment, allowing it to reinforce correct predictions and gradually improve its performance in an online learning framework [37, 38]. Therefore, while the RF model currently offers higher static prediction accuracy, the DNN model provides a scalable and dynamically tunable architecture that is well-suited for adaptive plasma process control when embedded within a feedback-driven reinforcement framework.

To further validate the effectiveness of the ML-based optimization framework, the predicted input parameter sets from both the DNN and RF models were experimentally tested. The corresponding OH(A-X) emission intensities observed under these optimal conditions showed strong agreement with the model-predicted values, confirming the physical relevance of the learned mappings. To ensure that these conditions indeed represent optimal regimes, a comparative analysis was also carried out using selected experimental data

points from the training dataset, including configurations that yielded maximum, minimum, and randomly distributed OH intensities.

The experimentally verified optimal conditions aligned well with those predicted by both models, reinforcing the robustness of the optimization process. In contrast, deviations from these conditions, particularly in the minimum and randomly chosen cases, resulted in substantially lower OH emission, thereby highlighting the sensitivity of plasma chemistry to input parameters. Overall, the consistent alignment between predicted and experimental outcomes establishes a strong foundation for implementing these models in adaptive control and optimization frameworks for plasma-based applications.

4. Conclusion

In this work, we designed and developed a multi-jet argon-based nonthermal plasma reactor and conducted systematic OES measurements under varying operating parameters, including gas flow rate, voltage, duty cycle, and % air. The resulting dataset, comprising 1500 well-labeled OES spectra, was used to investigate the feasibility of ML models in predicting emission characteristics of the multi-jet plasma system. PCA was employed solely as a dimensionality reduction tool to compress the high-resolution spectral data while preserving critical variance, and not as a learning model.

Two supervised learning models, RF and DNN, were trained to map experimental input parameters to key spectral features such as OH(A–X), Ar I, O-777 nm, and N₂ (C–B) emissions. The RF model demonstrated strong predictive accuracy, low training time, and robustness with limited data, making it highly suitable for real-time applications. The DNN model, although more sensitive to data sparsity, effectively captured complex nonlinear relationships and showed promise for future adaptive control frameworks. To extend the applicability of these models, a global optimization strategy based on differential evolution was implemented to identify the most effective input conditions for maximizing targeted emission features, specifically, OH(A–X), thus enabling inverse modeling and parameter tuning. The optimized conditions predicted by both models were experimentally verified and found to be consistent with high-OH-emitting regimes, further reinforcing the reliability and physical relevance of the data-driven optimization approach.

These results demonstrate that ML not only facilitates forward prediction of plasma behavior but can also be used for automated optimization and control of reactor parameters. The integration of such models with real-time data acquisition and closed-loop feedback systems holds significant potential for intelligent plasma reactor design, particularly in applications like water purification, surface treatment, and atmospheric pressure plasma chemistry. Future work will focus on expanding the dataset, refining the models, and implementing real-time feedback mechanisms to realize autonomous plasma process control. In addition, when it comes to the generalizability of these models, they perform best on

the specific reactor and measurement setup used for training. The workflow can be applied to other systems, but each new setup typically needs some calibration data to adapt the model. Predictions outside the trained operating window should be treated cautiously and, where possible, reported with uncertainty.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgments

P S N S R Srikar gratefully acknowledges the Ministry of Human Resource Development (MHRD), Government of India, for supporting his work through the HTRA fellowship. Indhu Suresh expresses sincere gratitude to the Council of Scientific and Industrial Research (CSIR), New Delhi, for providing both junior and senior research fellowships. R K Gangwar extends heartfelt thanks to SERB-DST, New Delhi, Government of India, for financial assistance under Grant No. CRG/2023/002553, as well as to DAE-BRNS, India, for additional support through Grant No. 57/14/01/2023-BRNS(MSS)/12089. The authors also wish to thank IIT Tirupati and CAMOST for their financial support.

Conflict of interest

The authors have no conflicts to disclose

Author contributions

P S N S R Srikar  0000-0001-5631-3055

Conceptualization (equal), Data curation (equal), Formal analysis (equal), Investigation (equal), Methodology (equal), Software (equal), Validation (equal), Visualization (equal), Writing – original draft (equal)

Indhu Suresh  0000-0002-6503-4039

Conceptualization (equal), Data curation (equal), Formal analysis (equal), Investigation (equal), Methodology (equal), Software (equal), Validation (equal), Visualization (equal), Writing – original draft (equal)

R K Gangwar  0000-0002-6376-8012

Conceptualization (equal), Formal analysis (equal), Funding acquisition (equal), Investigation (equal), Methodology (equal), Project administration (equal), Resources (equal), Software (equal), Supervision (equal), Validation (equal), Writing – review & editing (equal)

References

- [1] Adamovich I et al 2022 The 2022 plasma roadmap: low temperature plasma science and technology *J. Phys. D: Appl. Phys.* **55** 373001
- [2] Thagard S M and Locke B R 2022 Special issue of plasma chemistry and plasma processing scale-up of plasma reactors for bio, chemical, environmental, materials and energy applications *Plasma Chem. Plasma Process* **42** 1–2
- [3] Komarzynie G and Aftyka M 2020 Operating problems of arc plasma reactors powered by AC/DC/AC converters *Appl. Sci.* **10** 3295
- [4] Xiang Q, Fan L, Li Y, Dong S, Li K and Bai Y 2020 A review on recent advances in plasma-activated water for food safety: current applications and future trends *Crit. Rev. Food Sci. Nutr.* **62** 2250–68
- [5] Srikar P S N S R, Suresh I and Gangwar R K 2024 Accelerated real-time plasma diagnostics: integrating argon collisional-radiative model with machine learning methods *Spectrochim. Acta B* **215** 106909
- [6] Suresh I, Srikar P S N S R and Gangwar R K 2024 Integration of ML methods with CR model-based optical diagnostic for the estimation of electron temperature in Ga laser produced plasma *Phys. Plasmas* **31** 113501
- [7] Gidon D, Pei X, Bonzanini A D, Graves D B and Mesbah A 2019 Machine learning for real-time diagnostics of cold atmospheric plasma sources *IEEE Trans. Radiat. Plasma Med. Sci.* **3** 597–605
- [8] Park J-H, Cho J-H, Yoon J-S and Song J-H 2021 Machine learning prediction of electron density and temperature from optical emission spectroscopy in nitrogen plasma *Coatings* **11** 1221
- [9] Carter J A, O'Brien L M, Harville T, Jones B T and Donati G L 2021 Machine learning tools to estimate the severity of matrix effects and predict analyte recovery in inductively coupled plasma optical emission spectrometry *Talanta* **223** 121665
- [10] Allabakshi S M, Srikar P S N S R, Gangwar R K and Maliyekkal S M 2022 Application of plasma-assisted advanced oxidation processes for removal of emerging contaminants in water *New Trends in Emerging Environmental Contaminants. Energy, Environment and Sustainability* ed S P Singh, A K Agarwal, T Gupta and S M Maliyekkal (Springer) pp 333–70
- [11] Sarker I H 2021 Machine learning: algorithms, real-world applications and research directions *SN Comput. Sci.* **2** 160
- [12] Jiang T, Gradus J L and Rosellini A J 2020 Supervised machine learning: a brief primer *Behav. Ther.* **51** 675–87
- [13] Greenacre M, Groenen P J F and Hastie T 2022 Principal component analysis *Nat. Rev. Methods Primers* **2** 100
- [14] Ljubicic M L, Madsen A, Juul A, Almstrup K and Johannsen T H 2021 The application of principal component analysis on clinical and biochemical parameters exemplified in children with congenital adrenal hyperplasia *Front. Endocrinol.* **12** 652888
- [15] Huang Z, Yu X, Zhu D and Michael H C 2024 Interlude: interactions between labeled and unlabeled data to enhance semi-supervised learning (arXiv:2403.10658)
- [16] Elhaik E 2022 Principal Component Analyses (PCA)-based findings in population genetic studies are highly biased and must be reevaluated *Sci. Rep.* **12** 14683
- [17] Peng D, Gui Z and Wu H 2024 Interpreting the curse of dimensionality from distance concentration and manifold effect (arXiv:2401.00422)
- [18] Shlens J 2014 A tutorial on principal component analysis (arXiv:1404.110)
- [19] Marzban C, Yurtsever U and Richman M 2024 Principal component analysis for equation discovery (arXiv:2401.04797)
- [20] Brereton R G 2022 Numerical introduction to principal components analysis *J. Chemometr.* **36** e3405
- [21] Stelzer F, Röhm A and Vicente R 2021 Deep neural networks using a single neuron: folded-in-time architecture using feedback-modulated delay loops *Nat. Commun.* **12** 5164
- [22] Sazli M H 2006 A brief review of feed-forward neural networks *Communications Faculty of Sciences University of Ankara Series A2-A3 Physical Sciences and Engineering* **50** (https://doi.org/10.1501/commua1-2_0000000026)
- [23] Boué L 2018 Deep learning for pedestrians: backpropagation in cnns (arXiv:1811.11987)
- [24] Kingma D P and Jimmy B 2017 Adam: a method for stochastic optimization (arXiv:1412.6980)
- [25] Dutta P, Paul S and Kumar A 2021 Chapter 25 - comparative analysis of various supervised machine learning techniques for diagnosis of covid-19 *Electronic Devices, Circuits and Systems for Biomedical Applications* ed S L Tripathi, V E Balas, S K Mohapatra, K B Prakash and J Nayak (Academic) pp 521–40
- [26] Mienye I D and Sun Y 2022 A survey of ensemble learning: concepts, algorithms, applications and prospects *IEEE Access* **10** 99129–49
- [27] Das S, Dey A, Pal A and Roy N 2015 Applications of artificial intelligence in machine learning: review and prospect *Int. J. Comput. Appl.* **115** 31–41
- [28] Acquaviva V 2023 *Machine Learning for Physics and Astronomy* (Princeton University Press)
- [29] Shinde P P and Shah S 2018 A review of machine learning and deep learning applications 2018 4th Int. Conf. on Computing Communication Control and Automation (ICCUBEA) (IEEE) pp 1–6
- [30] Ribeiro M H D M and dos Santos Coelho L 2020 Ensemble approach based on bagging, boosting and stacking for short-term prediction in agribusiness time series *Appl. Soft Comput.* **86** 105837
- [31] Stefan D, Giotis K, Invernizzi L, Höft H, Hassouni K, Prasanna S, Svarnas P, Lombardi G and Gazeli K 2024 Machine learning assisted optical diagnostics on a cylindrical surface dielectric barrier discharge *J. Phys. D: Appl. Phys.* **57** 455206
- [32] Shadmehr R, Angell D, Chou P B, Oehrlein G S and Jaffe R S 1992 Principal component analysis of optical emission spectroscopy and mass spectrometry: application to reactive ion etch process parameter estimation using neural networks *J. Electrochem. Soc.* **139** 907
- [33] Mašláni A and Sember V 2014 Emission spectroscopy of oh radical in water-argon arc plasma jet *J. Spectrosc.* **2014** 952138
- [34] Friedl R and Fantz U 2012 Spectral intensity of the N2 emission in argon low-pressure arc discharges for lighting purposes *New J. Phys.* **14** 043016
- [35] Pourali N, Lai K, Gregory J, Gong Y, Hessel V and Rebrov E V 2023 Study of plasma parameters and gas heating in the voltage range of nondischarge to full-discharge in a methane-fed dielectric barrier discharge *Plasma Process. Polym.* **20** 2200086
- [36] Salgotra R and Gandomi A H 2024 A novel multi-hybrid differential evolution algorithm for optimization of frame structures *Sci. Rep.* **14** 4877
- [37] Sutton R S and Barto A G 1998 Reinforcement learning: an introduction *IEEE Trans. Neural Netw.* **9** 1054
- [38] Silver D et al 2017 Mastering chess and shogi by self-play with a general reinforcement learning algorithm (arXiv:1712.01815)